

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: AI-Assisted Early Risk Stratification of Adult

Emergency Department Patients Using Routinely Collected Triage Data

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Statement of the Problem

Emergency departments need practical ways to identify higher-risk triage patients earlier using routinely collected information. The current AI-assisted triage tools show promise but are often limited by poor explainability, weak local validation and poor fit with real ED workflow. This pilot will explore whether a practical AI triage support tool can help clinicians identify patients needing earlier review while keeping final decisions with the triage nurse and clinical team.

Previous Work

Previous work has shown growing interest in the use of artificial intelligence in emergency medicine, particularly for triage support, clinical disposition prediction, admission prediction, patient-flow planning and resource allocation ^{1,2}. A scoping review of AI-driven across various sources such as MEDLINE, Embase, PsycINFO, Web of Science and Scopus that mentioned triaging suggest that these systems may support earlier risk recognition, improve consistency and assist clinicians in high-pressure emergency department environments, but they also raise

concerns about explainability, workflow fit, fairness, validation and clinical trust²⁻⁵.

Several studies have explored the use of machine learning models to predict triage level, hospitalisation, length of stay, admission and other critical outcomes using routinely collected emergency department data^{2,6-9}. These studies support the idea that structured triage variables, including vital signs and clinical observations, can help identify patients who may need earlier review or escalation. For example, machine learning approaches have been used to predict hospital admission from ED data^{8,9}, while interpretable models have been developed to support triage accuracy and address issues such as class imbalance⁷. More recent AI-based triage studies also show potential for predicting critical outcomes, although local validation remains essential before clinical use^{6,10,11}.

Natural language processing has also become increasingly relevant to emergency department triage. Narrative and systematic reviews suggest that presenting complaint data, triage notes and other free-text documentation may contain clinically useful information that is not always captured in structured fields^{1,12}. Studies using machine learning and natural language processing have explored prediction of clinical disposition and triage performance using both structured and unstructured triage information^{12,13}. This is relevant to the proposed project because early presenting complaint information could support a red-flag intake prompt or assist with identifying patients who need priority nurse review.

However, much of the AI triage literature focuses heavily on predictive performance rather than implementation in real ED workflows. Measures such as accuracy, model discrimination, sensitivity, specificity and calibration are important, but they do not fully determine whether a tool will be useful in practice. A model may perform well retrospectively but still fail clinically if staff do not trust it, if alerts arrive at the wrong time, if the required data are unavailable, or if the

system adds extra steps to an already busy workflow¹⁴⁻¹⁸.

Workflow and human factors literature therefore provides an important foundation for this project.

Boonstra and Laven found that AI can influence the work design of emergency department clinicians, meaning that the effect of AI extends beyond prediction and can reshape tasks, responsibilities and decision-making processes¹⁴. Salwei et al. further emphasised that clinical decision support tools must be integrated into the ED workflow rather than added as a separate burden^{17,18}. Their work is especially relevant because the proposed system must fit around triage nurses, clinicians, handoffs, waiting-room reassessment and documentation processes.

The local Caribbean emergency department context is also important. De Freitas et al. mapped patient flow in a Caribbean emergency department and showed that ED flow is shaped by staffing, layout, documentation, handovers, delays, resource availability and external departments¹⁹. This supports the systems-thinking approach used in this project. The proposed AI tool cannot be designed as if the emergency department is an idealised digital environment. It must account for the possibility of missing data, delayed documentation, staff workload, handoff points and patients waiting before clinician review.

This project builds on previous AI triage and clinical decision support work by focusing not only on whether a model can identify risk, but where it would fit in the ED workflow and how clinicians would act on its output. The proposed system is designed as an explainable AI-assisted triage support tool that uses routinely collected triage information to support earlier identification of higher-risk patients. It would not replace triage nurses, ED consultants or clinical judgement. Instead, it would provide workflow-aware prompts at selected points, including intake, triage data review, vital sign pattern checking, waiting-room reassessment and clinician handoff.

Zotero Week 3 Reference Library:

Literature Review

Artificial intelligence is increasingly being explored in emergency medicine because emergency departments require rapid decision-making, prioritisation and coordination under pressure ^{2,4,20}. AI applications in emergency medicine include triage support, risk prediction, clinical disposition prediction, resource allocation, imaging support and patient-flow planning ^{2,4,20}. In triage, these tools commonly use data such as presenting complaint, demographics, vital signs, triage notes and electronic health record variables to estimate risk or predict downstream outcomes ^{1,6-9,12,13,21}.

AI-driven triage has several potential benefits. It may help identify critically ill patients earlier, improve consistency across triage decisions, support prioritisation during overcrowding and assist clinicians who are managing multiple patients with incomplete information ^{2,6,20,21}. Sitthiprawiat et al. developed and internally validated an AI-based emergency triage model for predicting critical outcomes, showing how AI can be used to support early risk identification in the ED⁶. Lin et al. also developed an interpretable deep learning system for identifying critical patients by predicting triage level, hospitalisation and length of stay ²¹. These studies support the potential value of using routinely collected triage data to identify patients who may require earlier clinical review.

Machine learning has also been applied to admission and disposition prediction. Predicting hospital admission from the emergency department is relevant because it can support earlier planning, improve patient flow and help clinicians anticipate downstream resource needs. Patel et al. and Feretzakis et al. both explored machine learning approaches for predicting hospital admission from ED data ^{8,9}. Chang et al. used machine learning and natural language processing to

predict clinical disposition in the emergency department, further supporting the use of triage data for workflow-relevant decision support¹³. These studies are relevant to this project because they show that ED data can be used not only for triage scoring, but also to support decisions that affect patient flow and disposition.

Natural language processing is another important area of triage research. Triage notes and presenting complaints often contain information that may not be fully captured in structured fields. Stewart et al. reviewed applications of natural language processing at ED triage and highlighted its potential for extracting useful information from triage documentation¹. Porto also reviewed the use of machine learning and natural language processing to improve triage performance, showing that unstructured clinical text can contribute to triage support when used appropriately¹². This supports the inclusion of a red-flag intake prompt in the proposed system because early presenting complaint information may help identify patients who need priority nurse review. However, NLP-based tools must be designed carefully because language, documentation habits, spelling variation and missing information can affect performance.

Despite these promising findings, the literature also shows that predictive performance alone is not enough. AI tools must be understandable, clinically useful and integrated into the workflow. Explainability is especially important in emergency medicine because clinicians need to understand why a patient has been flagged, particularly when decisions may affect waiting time, escalation or allocation of urgent care. Okada et al. emphasised the importance of explainable AI in emergency medicine²². Rosenbacke et al. also showed that explainable AI can either increase or decrease clinician trust depending on how explanations are designed and presented²³. Perivolaris et al. further highlighted that clinician-AI interaction is shaped by trust, usefulness, usability, satisfaction and willingness to use the system¹⁵. These findings support the design of a vital sign pattern alert and clinician handoff summary rather than a standalone unexplained score.

Workflow integration is another major theme. Boonstra and Laven showed that AI can influence the work design of emergency department clinicians, meaning that AI systems may change tasks, roles and decision-making processes¹⁴. This is important because an AI triage tool is not neutral; it affects nurses, doctors, administrative staff, support staff, patients and clinical IT teams. Salwei et al. analysed workflow integration of a human factors-based clinical decision support tool in the ED and emphasised the need to fit clinical decision support into existing work processes¹⁸. Their later work on usability also showed that design and implementation lessons are critical for clinical decision support tools in emergency settings¹⁷. These studies support the Week 3 focus on diagramming the workflow before designing or deploying the AI system.

Nursing adoption is particularly relevant because the triage nurse is a primary user of the proposed system. Zhai et al. identified barriers and facilitators to implementing a nursing clinical decision support system using the FITT framework, which considers the fit between individuals, tasks and technology²⁴. This is directly relevant to the proposed AI tool because it must support triage nurses without increasing workload, undermining professional judgement or creating unnecessary documentation. A system that requires too many clicks, opens on a separate screen or produces unclear alerts may increase friction rather than improve care.

Alert fatigue is another major implementation risk. Olakotan and Yusof found that clinical decision support alerts must be appropriate to the clinical workflow to be useful¹⁶. If alerts are too frequent, poorly timed or clinically irrelevant, clinicians may override or ignore them. This is a significant concern for an AI-assisted triage system because excessive alerts could reduce trust and contribute to missed warnings. Therefore, the proposed system should use limited, explainable and high-value prompts. It should not generate alerts for every minor abnormality, and it should clearly show why a patient was flagged.

The local ED system also matters. De Freitas et al. conducted a qualitative exploration of patient

flow in a Caribbean emergency department and showed that emergency care is shaped by practical system factors such as staffing, layout, documentation processes, handovers, delays and external departments ¹⁹. This finding is important because many AI triage models are developed in settings that may not reflect Caribbean emergency department realities. If an AI model assumes complete real-time electronic data, but the actual workflow includes delayed entry or paper-first documentation, the model may fail at the exact moment it is needed. This supports the proposed missing-data safety check, which would warn clinicians when key triage fields are incomplete before a risk flag is interpreted.

Ethical considerations are also central to emergency triage. Cha and Kim discuss ethical issues related to AI in emergency medicine triage and resource allocation, including fairness, transparency, accountability and the risk of harm if AI systems influence prioritisation decisions inappropriately ⁵. Araouchi and Adda also present an AI-assisted multimodal triage system for health centres, showing broader interest in AI triage tools while reinforcing the need to consider implementation context²⁵. In emergency care, AI outputs may affect who is seen sooner, who waits, and who receives earlier escalation. For this reason, human oversight must remain central. The proposed system should keep final decisions with triage nurses and clinicians, while using AI as an additional support layer.

The literature therefore supports the potential value of AI-assisted triage, but also shows that technical performance alone is insufficient. AI tools must be locally validated, explainable, usable, ethically safe and designed around the real ED workflow ^{5,14–20,22,23}. This project responds to those findings by proposing a workflow-aware AI triage support system that uses routinely collected triage information to identify higher-risk patients earlier while preserving clinician judgment. The system is intended to plug into specific workflow points: intake, triage data review, vital sign pattern checking, waiting-room reassessment and clinician handoff. By focusing on both

prediction and workflow fit, the project aims to develop a more practical and responsible approach to AI-assisted emergency department triage.

Proposed Solution

My proposed solution is a 12-week pilot of a flow-aware, explainable triage risk support tool for adult emergency department patients at Mercer General. The tool would use routinely collected information from the early patient journey to support faster recognition of patients who may be at higher risk of deterioration, urgent clinical need or delay within the emergency department workflow.

The tool would not replace triage nurses or automate final triage decisions. Instead, it would support the triage process by helping staff collect relevant information earlier, identify concerning clinical patterns and explain why a patient may need prompt review. The final triage decision would remain with the triage nurse and clinical team.

A key feature of the proposed tool is that it would have three connected workflow layers.

Layer 1: Reception-assisted digital intake

In many emergency departments, the first interaction may happen before formal nurse triage. A receptionist or intake staff member could enter basic non-clinical information into the tool, such as age, sex or gender, arrival mode and presenting complaint. The intake screen could also include a structured red-flag checklist for obvious urgent symptoms such as chest pain, shortness of breath, fainting, severe bleeding, confusion or visible distress.

This step would not allow reception staff to make clinical decisions. Instead, it would help

capture early information and prompt faster nurse review when a red-flag symptom is reported. For example, if a patient reports shortness of breath or chest pain at registration, the tool could display: “Priority nurse review recommended.” The triage nurse would still be responsible for confirming the clinical concern, recording or validating vital signs and making the final triage decision.

Layer 2: Explainable clinical risk stratification

After the triage nurse records routine clinical observations, the tool would use routinely collected triage variables to assign the patient to a low, moderate or high-risk category. These variables could include:

Age

Gender

Systolic blood pressure

Diastolic blood pressure

Mean arterial pressure

Pulse

Temperature

Respiratory rate

GCS

SpO₂ where available

FiO₂ where available

Presenting complaint where available

The tool would not simply produce a risk score. It would also provide a short explanation in plain

clinical language so that the nurse can understand why the patient was flagged. For example:

“High risk: low MAP and high pulse detected. Review for possible poor perfusion or clinical deterioration.”

This explainability is important because a nurse-facing tool must be understandable at the bedside. A risk category alone may not be enough if the user cannot see the clinical reasons behind it.

Layer 3: Flow-aware escalation support

The third layer would make the tool more specific to the realities of emergency department workflow. It would consider whether a patient who appears clinically high risk may also be vulnerable to delays in the ED system. For example, a patient who is high risk and likely to require oxygen support, imaging, admission, senior clinical review or close monitoring may need earlier escalation to avoid deterioration while waiting.

This layer is important because emergency department risk is not only about the patient’s vital signs. It is also about whether the patient may become delayed at a point in the care pathway. In this way, the tool would ask two practical questions:

1. Is this patient clinically high risk?
2. Is this high-risk patient also likely to be affected by delays in the ED workflow?

This three-part thinking makes the proposal more useful for Mercer General because it connects clinical risk recognition with patient flow awareness.

The pilot could be structured as follows:

Timeline	Deliverable
Weeks 1 to 2	Confirm the clinical problem, review the current triage and registration workflow and identify what information is collected at each stage.
Weeks 3 to 4	Prepare and clean the triage dataset, including checking missing values, validating vital sign ranges and documenting available variables.
Weeks 5 to 6	Build a first prototype using routine triage variables and simple risk rules or a basic machine learning model.
Weeks 7 to 8	Add the explanation layer so that each risk output includes clear clinical reasons for the flag.
Weeks 9 to 10	Test the tool on retrospective triage data to check whether the risk categories and explanations make clinical sense.

Weeks 11 to 12	Collect feedback from nurses or clinical reviewers on whether the tool is clear, useful and realistic for triage and refine the prototype based on that feedback.
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This proposal is feasible because it uses information already collected during the early ED journey and keeps the nurse at the centre of decision-making. It also addresses the main gaps identified in the literature: the need for explainable nurse-facing decision support, practical testing using early triage data and better alignment with local patient flow challenges.

Methodology of Workflow Mapping

The purpose of this Week 3 interim submission is to map the current emergency department triage workflow and identify where an AI-assisted triage support system could realistically fit. This stage is important because a model that performs well in isolation may still fail if it does not match the clinical workflow, staffing realities, data-entry process, and delays of the emergency department.

For this project, the AI system is treated as one component within a larger clinical system. The workflow therefore, considers inputs, outputs, actors, constraints and feedback loops. The aim is not only to ask whether an AI model can identify higher-risk patients, but whether it can do so in a way that reduces friction for clinical staff and supports safer decision-making in the actual ED environment.

The workflow is informed by De Freitas et al., who used qualitative observational process mapping to study patient flow in a Caribbean emergency department. Their study is especially

relevant because it highlights that ED flow is shaped by work processes, layout, staffing, delays, resources and external departments, rather than by clinical decisions alone¹⁹. This is important for Mercer General because an AI triage support system must be designed for the workflow that exists, not an idealised version of emergency care.

ED Triage Workflow Notes

The ED triage workflow begins when a patient arrives in the emergency department. The patient may arrive by ambulance, walk in independently, be accompanied by a relative or caregiver, or be referred from another health facility. At arrival, staff must first determine whether the patient is critically unwell and needs immediate resuscitation or whether the patient can proceed through the standard registration and triage pathway.

If the patient appears critically unwell, the patient should be directed immediately to the resuscitation room for urgent clinician review, stabilisation and investigations. This route must remain separate from any routine AI-assisted scoring process because immediate clinical action should not be delayed by digital intake, registration or model output.

If the patient is not critically unwell, the next decision is whether the emergency department is the appropriate place for care. Some patients may be redirected to a local health centre if their condition does not require ED-level care. Patients who remain in the ED proceed to registration or intake, where demographic details and the presenting complaint are captured. This is a key early point where information enters the system, but it is also a potential delay point if registration and triage are physically or operationally separated.

After registration, the patient waits for the triage nurse. The triage nurse then assesses the patient and records routine triage information such as presenting complaint, pulse, blood pressure,

respiratory rate, temperature, oxygen saturation, GCS and other relevant observations. This is the main point where the proposed AI-assisted system could use routinely collected triage data to support earlier identification of higher-risk patients.

However, ED workflows may include delays and workarounds, including information being recorded before it becomes available in the formal electronic system¹. This means an AI plug-in cannot assume that all information is immediately available. If vital signs or presenting complaints are missing, delayed or inconsistently entered, the system should recognise that limitation rather than produce a misleading risk flag.

Once triage data is available, the triage nurse or doctor assigns a triage category. At this stage, the AI system could support the clinician by flagging concerning patterns and explaining why the patient may need faster review. The tool should not replace the triage nurse's judgement or assign final priority independently. Instead, it should assist the nurse by making risk information easier to interpret.

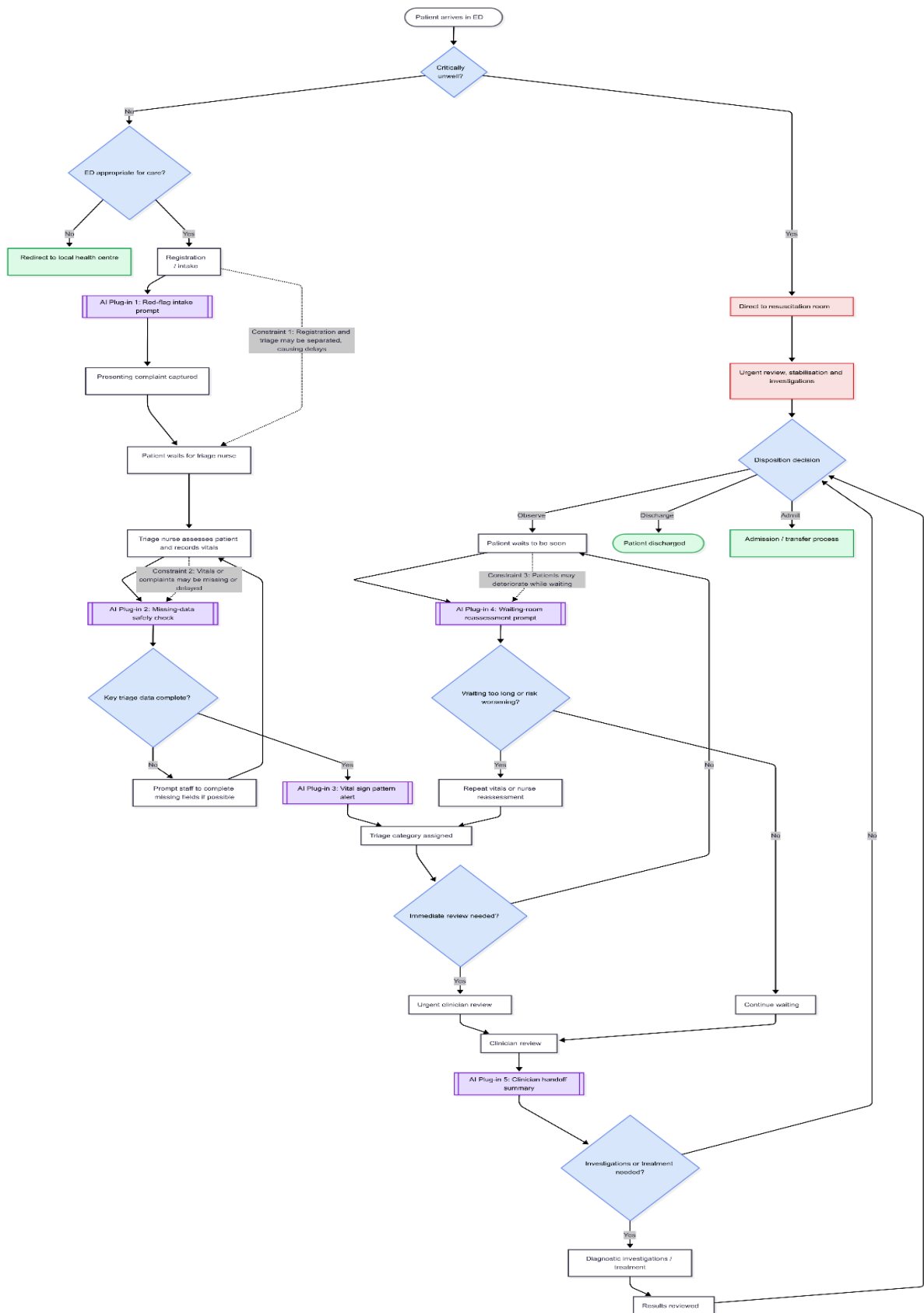
After triage, the patient may receive urgent clinician review, wait to be seen, undergo diagnostic investigations or treatment, be observed, discharged, admitted or transferred. Waiting is a major part of the ED process. A patient may wait after registration, after triage, before clinician review, while awaiting investigations, while awaiting results or while awaiting admission or transfer. Because risk can change during waiting, a one-time triage flag may not be enough. A useful system should help identify when reassessment or repeat vital signs may be needed.

Workflow Diagram

The workflow diagram was created using Mermaid. The diagram shows the patient journey from ED arrival through critical assessment, registration, triage, waiting, clinician review, investigations

or treatment and disposition. It also includes five AI plug-in points and three candidate workflow constraints.

Figure 1. ED triage workflow diagram with AI plug-in points and workflow constraints.



The diagram is intended to remain readable for a clinical reviewer. It therefore simplifies the full patient flow process into the core steps most relevant to the proposed AI-assisted triage support system. The Mermaid source file and exported diagram image will be committed to the GitHub repository so that the diagram is both readable and version-controlled.

See workflow diagram within /plots in Github repo:

<https://github.com/ShariUWI/carisurg-portfolio/tree/main/plots>

AI Plug-In Points

The AI plug-in points were selected based on where information enters the ED workflow, where decisions are made, where delays occur and where risk information may be lost. For each plug-in point, the key question is: what goes in, what comes out and what does the human do next?

AI Plug-In 1: Red-Flag Intake Prompt

Workflow location: Registration or intake

Input: Presenting complaint and obvious red-flag symptoms

Output: Prompt for priority nurse review if concerning symptoms are reported

Human action: Registration or intake staff notify the triage nurse without making a clinical decision

This plug-in is intended to support early recognition of symptoms such as chest pain, shortness of breath, fainting, confusion, severe bleeding or visible distress. It does not allow non-clinical staff to triage the patient. Instead, it supports the handoff from intake to the triage nurse. This is relevant because machine learning and natural language processing methods can support triage

performance when presenting complaint information and other triage data are used appropriately¹².

Upstream: Patient has arrived and is being registered.

Downstream: Triage nurse is prompted to review the patient earlier if red-flag symptoms are present.

AI Plug-In 2: Missing-Data Safety Check

Workflow location: After triage data entry

Input: Routine triage fields such as presenting complaint, pulse, blood pressure, respiratory rate, temperature, oxygen saturation and GCS

Output: Prompt showing which key fields are missing or delayed

Human action: Staff complete missing fields where clinically possible or proceed with awareness that the risk assessment is incomplete

This plug-in is included because AI systems are only as useful as the data available at the time they are used. If important fields are missing, the system should not produce a confident risk flag without warning the clinician. This addresses the common workflow problem of delayed, incomplete or inconsistent data entry. Poor workflow fit and additional documentation burden are known barriers to clinical decision support adoption^{14,18}.

Upstream: The triage nurse has assessed the patient and entered available information.

Downstream: The system either confirms sufficient data for risk support or prompts staff to complete missing fields.

AI Plug-In 3: Vital Sign Pattern Alert

Workflow location: After key triage data are complete

Input: Vital signs and presenting complaint

Output: Plain risk alert identifying concerning clinical patterns

Human action: Triage nurse compares the alert with their clinical judgement and decides whether to escalate, monitor or continue routine triage

This plug-in supports the core project aim: earlier identification of higher-risk ED patients using routinely collected triage information. The tool should avoid presenting only a raw score. Instead, it should explain the reason for concern, such as low oxygen saturation, high respiratory rate, abnormal blood pressure or reduced GCS. Clinician trust in AI depends not only on accuracy, but also on whether the output is understandable and usable in clinical work⁴.

Upstream: Triage data have been captured and checked for completeness.

Downstream: The triage nurse uses the alert as decision support before assigning or confirming triage priority.

AI Plug-In 4: Waiting-Room Reassessment Prompt

Workflow location: Patient waiting after triage

Input: Time since triage, risk flag status and whether repeat observations have been recorded

Output: Prompt for repeat vital signs or nurse reassessment if a higher-risk patient waits too long

Human action: Nurse reassesses the patient or repeats vital signs if clinically appropriate

This plug-in is included because ED risk is not static. A patient may appear stable at triage but deteriorate while waiting. Patient waiting and flow delays are prominent features of ED patient journeys, especially when staffing, bed availability or investigations create downstream

bottlenecks¹. A reassessment prompt supports the workflow without assuming that the initial triage decision remains valid throughout the ED stay.

Upstream: Patient has been triaged and is waiting to be seen.

Downstream: If risk worsens or waiting time becomes concerning, repeat observations or reassessment are triggered.

AI Plug-In 5: Clinician Handoff Summary

Workflow location: Before or during clinician review

Input: Presenting complaint, relevant vital signs, risk flag and reason for flag

Output: Short plain-language summary of why the patient was flagged

Human action: Clinician reviews the summary alongside the patient record and clinical assessment

This plug-in is designed to reduce information loss between triage and clinician review. Handoffs are high-risk points because information can be delayed, simplified or lost. A short handoff summary can preserve the reason for an earlier risk flag and help clinicians understand why the patient may need attention. This supports explainability and clinical defensibility while keeping the clinician responsible for final decisions.

Upstream: Triage assessment and risk flag have already occurred.

Downstream: Clinician receives a concise explanation at review, rather than only a score or alert.

Candidate Workflow Constraints

Constraint 1: Registration and Triage May Be Separated, Causing Delays

The workflow may include a gap between registration and triage. During this gap, important information such as presenting complaints or visible distress may be captured informally before the patient is formally assessed by a triage nurse. De Freitas et al. identified layout, work processes and staffing as important factors influencing ED flow in a Caribbean setting¹⁹.

Design implication:

The AI system should not assume that complete clinical data are available at arrival. Early plug-ins should use only data realistically available at that point, such as presenting complaints or red-flag symptoms and should route concerns to the triage nurse rather than make clinical decisions independently.

Constraint 2: Vital Signs or Presenting Complaints May Be Missing, Delayed or Inconsistently Entered

Triage data may not be immediately available in a complete digital format. Some information may be captured verbally or on paper before it is entered into an electronic system. If an AI tool depends on complete real-time data, it may fail or produce unreliable output when fields are missing.

Design implication:

The system should include a missing-data safety check before producing a risk flag. It should clearly show when the risk assessment is limited by missing information. This reduces the risk of false reassurance and supports safer use of AI in a real clinical environment.

Constraint 3: Patients May Deteriorate While Waiting

The ED workflow includes multiple waiting points: waiting for triage, waiting for clinician review, waiting for investigations, waiting for results and waiting for admission or transfer. If the AI tool only produces one static alert at triage, it may miss deterioration that occurs later in the patient journey.

Design implication:

The system should support reassessment prompts for higher-risk patients who wait too long or have not had repeat observations. This allows the AI tool to support patient flow and safety beyond the first triage decision.

Constraint 4: Alert Fatigue and Poor Trust May Lead Clinicians to Ignore the System

If the system produces too many alerts, unclear alerts or alerts that do not fit the workflow, clinicians may ignore or override them. Alert fatigue is a recognised issue in clinical decision support systems, and inappropriate alerts can interrupt clinical work rather than improve it²⁰.

Design implication:

The system should prioritise high-value alerts, explain why the patient was flagged and avoid creating unnecessary steps for triage nurses. The tool should reduce cognitive load at a bottleneck rather than add work.

Stakeholder Considerations

This proposal includes a developed stakeholder map, the following stakeholders have already been identified because each may accept, reject or reshape the proposed system.

Stakeholder	Main Concern	What Success Looks Like
Triage nurse	The system must support speed and accuracy without adding extra steps during a busy shift.	The nurse sees a clear, useful alert that supports their judgement and does not slow triage.
ED consultant or doctor	The system must provide clinically defensible reasoning, not unexplained scores.	The clinician can understand why a patient was flagged and decide whether escalation is justified.
Registration or intake staff	The system must not require non-clinical staff to make clinical decisions.	Staff can enter presenting complaint information and red-flag symptoms without being asked to triage.
Porter or support staff	The system should not create extra movement, scanning or documentation tasks.	The tool does not depend on support staff completing unfamiliar digital steps before patient care moves forward.

Patient or patient advocate	The system must be fair, transparent and not delay care for patients who present calmly but are clinically at risk.	Patients benefit from earlier recognition of risk without being disadvantaged by missing data, language barriers or biased assumptions.
Clinical IT or EHR lead	The system must be secure, auditable, reliable and able to fit existing documentation processes.	The tool can be reviewed, monitored and safely integrated without exposing private data or creating unreliable workflow dependencies.

This stakeholder mapping reflects the idea that technical performance alone is not enough. A clinically useful AI tool must fit the needs of the people who act on it. If the triage nurse ignores the alert, the tool has no clinical value even if the model performs well in retrospective testing.

Friction Points

The following friction points were identified from the workflow diagram and workflow notes:

- 1. Data may not exist at the moment the AI is expected to fire.**

For example, a risk model cannot use complete vital signs if vital signs have not yet been taken or entered.

2. **The triage nurse may not have time to interact with a separate tool.**

A tool that requires extra screens, forms or manual confirmation may fail during high-pressure periods.

3. **Alerts may become noise if they are too frequent or poorly explained.**

This could reduce trust and cause clinicians to ignore the system.

4. **Handoff points may lose important information.**

A patient flagged during triage may later be reviewed by another clinician who does not understand why the patient was flagged.

5. **Waiting time changes risk.**

A patient's status may worsen while waiting, so one-time triage scoring may not be sufficient.

6. **Local workflow realities may differ from international AI literature.**

ED flow in Caribbean settings may involve different staffing, layout, infrastructure, patient mix and resource constraints from the settings where many AI tools are developed or tested ¹

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